

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 03: E-Commerce Data Foundations: Events, Experimentation, and Metrics

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Session plan (90 minutes)

- 1 Why data foundations matter
- 2 From business process to event model
- 3 Event schema design
- 4 Identity resolution and entity modeling
- 5 Data quality as a product requirement
- 6 Hands-on lab: build a minimal metrics layer

Learning objectives

By the end of this lecture, you should be able to:

- Design an event-driven data model for storefront, marketplace, and mobile commerce scenarios.
- Distinguish identifiers, identities, sessions, and entities across operational and analytical systems.
- Explain how attribution, data quality, and metric definitions affect decision-making and ML performance.
- Plan controlled experiments and understand when adaptive approaches such as bandits are appropriate.

Why data foundations matter

- Product analytics and funnel analysis
- Operational reporting across commerce and ERP systems
- Reliable feature generation for ML models
- Controlled online experimentation
- Governance, auditing, and cross-functional alignment

From business process to event model

- **Events versus state:** **State** answers "what is true now?" Examples: current inventory, current product price, current loyalty tier.; **Events** answer "what happened?" Examples: product viewed, cart updated, payment failed, order shipped.
- **Event taxonomies:** **Discovery events:** homepage viewed, category viewed, search submitted, filter applied, result clicked.; **Consideration events:** product viewed, review expanded, comparison opened, wishlist updated.

- **Required design principles: Business semantics before UI semantics:** schemas should reflect domain meaning, not implementation accidents.; **Stable field names:** renaming fields casually breaks dashboards, pipelines, and models.
- **Example event record:** The following conceptual record shows how a product view may be represented: The example illustrates an important practice: authenticated and anonymous identifiers may coexist.

Identity resolution and entity modeling

- **Identifiers are not interchangeable:** **User/account ID:** application-level identity for authentication and personalization.; **Customer/partner ID:** commercial identity used in ERP, billing, and support.
- **Identity stitching:** linking a guest cart to a logged-in account after authentication,; attaching a mobile device history to a user after sign-in,
- **Privacy-aware identity design:** minimizing collection of directly identifying information,; separating analytical IDs from operational personal data where possible,

Data quality as a product requirement

- **Data contracts:** event name and schema,; field definitions and allowed values,
- **Validation and observability:** row-count anomalies by event family,; null-rate spikes in required fields,

Attribution and its limits

- **Common attribution models:** **Last-touch attribution:** credits the final recorded touchpoint before conversion.; **First-touch attribution:** credits the earliest touchpoint in the conversion path.
- **Selection bias and measurement bias:** **Selection bias:** users exposed to an experience may differ systematically from users who are not.; **Survivorship bias:** only recorded users or channels appear in the analysis.

Metrics layers and decision quality

- **North Star and supporting metrics:** traffic volume,; product-detail engagement,
- **Metric dimensions:** channel (web, app, store-assisted, marketplace),; customer segment (new, returning, VIP, B2B account tier),
- **Retention and cohort logic:** Did the user return to browse?; Did the user make another purchase?

Experimentation in e-commerce

- **A/B testing fundamentals:** **Unit of randomization:** user, session, device, seller, or region.; **Exposure logic:** where and when eligibility is checked.
- **Common experimental pitfalls:** **Contamination:** the same user sees both variants through multiple devices or sessions.; **Novelty effects:** short-term excitement masks weak long-term performance.
- **Bandits and adaptive experimentation:** promotions on fast-moving catalogs,; ranking widgets with many interchangeable treatments,

From analytics to machine learning features

- recent category views for recommendation,
- failed login or payment attempts for fraud detection,
- coupon usage history for promotion targeting,
- time since last order for churn models,
- return behavior and order defects for seller-quality models.

Hands-on lab: build a minimal metrics layer

- **Lab scenario:** Assume a storefront emits five core events: , , , , and .
- **Lab tasks:** Create a canonical event table with required fields, event validation flags, and ingestion metadata.; Build daily metrics for sessions, product views, add-to-cart rate, checkout-start rate, purchase conversion rate, and day-7 repeat purchase.
- **Expected learning outcome:** By the end of the lab, students should be able to explain why a metric is not just an aggregation, but the result of event design choices, identity assumptions, and validation rules.

Managerial and architectural implications

- Which identifiers are authoritative, and where are they mastered?
- Which events belong in the clickstream versus operational event streams?
- How are storefront events linked to ERP objects such as orders, invoices, and shipments?
- What happens when a schema changes in the storefront while ERP integrations remain stable, or vice versa?
- Who approves metric definitions that drive executive dashboards and model targets?